

Integration and Summation

1 Integration

- symbolic and numeric integration
- symbolic bounds and singularities

2 Summation

- explicit formulas for sums
- limits of sums

MCS 320 Lecture 23
Introduction to Symbolic Computation
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Symbolic and Numeric Integration

Consider an expression $f(x)$ in a variable x .

If a symbolic antiderivative for $f(x)$ exists, then

$$\frac{d}{dx} \int f(x) dx = f(x) = \int \frac{d}{dx} f(x) dx.$$

In many cases, there is no symbolic antiderivative and we have to numerically approximate the definite integral

$$\int_a^b f(x) dx.$$

Theorem (Fundamental Theorem of Calculus)

Let $F(x)$ be the antiderivative of $f(x)$. Then,

$$\int_a^b f(x) dx = F(b) - F(a).$$

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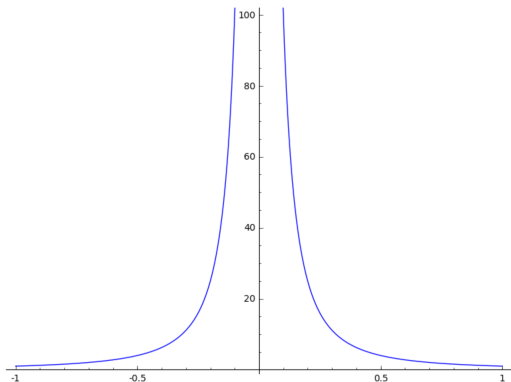
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Symbolic Bounds and Singularities

$$\int_a^b \frac{1}{x^2} dx = \left. \frac{-1}{x} \right|_{x=a}^{x=b} = \frac{-1}{b} - \frac{-1}{a}$$

The above formal calculation does not work for $[a, b] = [-1, +1]$:



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Explicit Formulas for Sums

Consider the sum of the first n positive integers:

$$\begin{aligned}\sum_{k=1}^n k &= 1 + 2 + 3 + \cdots + n \\ &= \frac{1}{2}n(n+1).\end{aligned}$$

The geometric sum for the ratio $r \neq 1$ is

$$\begin{aligned}\sum_{k=0}^n r^k &= 1 + r + r^2 + \cdots + r^n \\ &= \frac{1 - r^{n+1}}{1 - r}.\end{aligned}$$

In the limit, as $r \rightarrow 1$, the sum equals $n + 1$.

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Limits of Sums

The sum with infinitely many terms

$$\sum_{k=1}^{\infty} \frac{1}{k^2}$$

is equivalent to

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{k^2}$$

which equals

$$\frac{1}{6}\pi^2.$$